LOT Pager — Components Buying List (BOM)

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Prices are planning estimates for 100-unit quantity (Nov 2026), USD, not quotes. Links are distributor **search** links so they don't break when a listing changes. Check stock and lifecycle status before ordering. Items marked **VERIFY** need a datasheet check of size or part number before layout.

A. Stage 0 — breadboard kit (buy now, 1–2 sets)

Proves the whole loop (notice → screen → Copy → Log tab) before any custom PCB.

Qty	Item	Why	Est.	Link
1	Seeed Studio XIAO ESP32S3 **Sense** (camera + PSRAM)	Same MCU family as final; has a camera	$14	DigiKey
1	Arduino Nicla Sense ME (BME688 + BHI260AP on one board)	Exactly the final AI sensors	$75	DigiKey
1	Adafruit SHARP Memory Display breakout 1.3" (#3502)	Same display tech as final	$25	Adafruit
1	Adafruit DRV2605L haptic driver (#2305) + LRA motor (#1640 ERM or any LRA)	Pager buzz	$13	Adafruit
1	Qi receiver module, 5 V, ~30 mm coil (generic) + Qi pad	Prove charging through glass vs. steel	$15	DigiKey
1	LiPo 3.7 V ~150 mAh with protection	Battery-life measurements	$7	Adafruit
1	Tactile button, jumper wires, breadboard	Copy button	$8	—
1	316L stainless sheet sample, 0.5 mm, 50 × 50 mm	Qi + Wi-Fi attenuation test	$10	McMaster
	Stage 0 total		**~$167**	

B. Production BOM (per unit, 100 units)

Ref	Part	Function	Package / size	Est. @100	Link
U1	Espressif **ESP32-S3-PICO-1-N8R8**	MCU, Wi-Fi 4 + BLE 5, 8 MB flash + 8 MB PSRAM, camera DVP	LGA 7 × 7 mm SiP	4.80	DigiKey · LCSC
ANT1	Johanson 2450AT18A100 (or equal)	2.4 GHz chip antenna, under glass window	0603-size	0.45	DigiKey
Y1	40 MHz crystal	ESP32-S3 clock (check SiP — some PICO variants integrate it) **VERIFY**	2016	0.30	LCSC
Y2	32.768 kHz crystal	Low-power RTC for scheduled wake	1610	0.30	LCSC
U2	Bosch **BME688**	Weather: temp, humidity, pressure, gas (VOC) — AI-trainable via BME AI-Studio	LGA 3 × 3 × 0.93 mm	8.50	DigiKey
U3	Bosch **BHI260AP**	Self-learning AI motion sensor: pick-up wake, tap, gesture	LGA 3.6 × 4.1 mm	7.00	DigiKey
U4	Himax **HM0360** camera module (VGA, ultra-low-power, AI vision)	Camera	module ≤ 2.5 mm Z **VERIFY**	11.00	DigiKey

Ref	Part	Function	Package / size	Est. @100	Link
LCD1	Sharp **LS013B7DH03** 1.28" 128 × 128 memory LCD	Silver reflective, µW static power, readable in sun	~0.7–1.0 mm panel **VERIFY**	13.50	DigiKey
U5	TI **BQ51050B**	Qi receiver + Li-ion charger in one IC	DSBGA / VQFN	3.20	DigiKey
L1	Qi receiver coil, ~30 mm, ≤ 0.6 mm thick (Würth WE-WPCC family or equal) **VERIFY**	Wireless charging	flat coil	3.00	DigiKey
FS1	Ferrite shielding sheet 30 × 30 × 0.1 mm	Shields battery/PCB from Qi field	sheet	0.60	DigiKey
BT1	LiPo cell, 1.5 mm × 30 × 30 mm class, ~80 mAh, with PCM	Battery (semi-custom; MOQ often 500+, ask PCBWay/vendor)	pouch	4.50	search
U6	Analog Devices **MAX17048**	Fuel gauge (battery %)	WLP 0.9 × 1.7 mm	1.40	DigiKey
U7	TI **TPS62840**	3.0 V buck, 60 nA quiescent	WCSP	1.10	DigiKey
U8	Microchip **ATECC608B**	Secure device identity / token storage for LOT API	UDFN 2 × 3 mm	0.80	DigiKey
U9	TI **DRV2605L**	Haptic driver (pager buzz patterns)	WCSP / VSSOP	1.60	DigiKey
M1	Coin LRA, ≤ 8 mm Ø × ≤ 2.5 mm	Vibration	coin	1.50	DigiKey
U10	Vishay **VEML6030**	Ambient light → auto screen frontlight off/on, context	2 × 2 mm	0.90	DigiKey
SW1	Metal snap dome, 4 mm, ~250 gf	Copy button under steel key	0.3 mm	0.10	DigiKey
J1, J2	0.3 mm pitch FPC connectors (LCD, camera)		low-profile ≤ 0.6 mm	1.20	DigiKey
VENT1	ePTFE adhesive vent (Gore PolyVent class), Ø 3 mm	Lets BME688 breathe, keeps water out	0.2 mm	1.20	DigiKey
—	Passives (~60 × 0201/0402), ESD diodes			2.00	—
—	Factory pogo pads (UART, EN, BOOT, GND, VBAT)	Flash + test; no USB port on device (sealed)	pads only	0.00	—
	Electronics subtotal			**68.95**	
	+ 5 % attrition / spares			**≈ 72.40**	

C. Mechanical and packaging (per unit)

Item	Spec	Vendor	Est. @100
Back shell (Part A)	316L, CNC, #8 mirror polish outside, bead-blast inside	PCBWay CNC	22.00
Front frame (Part B)	316L, CNC, brushed or polished, window pocket + key hole	PCBWay CNC	16.00
Steel key cap	316L, 5 mm Ø, laser-etched "COPY"	PCBWay CNC	(incl.)
Cover glass	0.55 mm aluminosilicate (or 0.4 mm sapphire, +$8), AR coat, black print mask with camera hole	glass vendor via PCBWay	6.00
Screws / adhesive	4 × M1.2 stainless Torx or 3M VHB ring	McMaster / PCBWay	0.60

Item	Spec	Vendor	Est. @100
Qi charging pad (5 W, off-the-shelf, white-label)	in box	Alibaba / Anker class	7.50
USB-C cable + box + printed quick-start		local printer	5.90

D. Where to buy

- **PCBWay** — PCB, turnkey PCBA (they source parts), CNC stainless, polishing: https://www.pcbway.com
- **DigiKey / Mouser** — sensors, ICs, dev kits (best for Stage 0 and for consigning hard-to-find ICs to PCBWay)
- **LCSC** — passives, crystals, connectors at low cost (PCBWay can source from here)
- **Bosch Sensortec** — BME AI-Studio (free) for training gas/odour classes on BME688

E. Alternatives (if a part is short or too thick)

Primary	Alternative	Trade-off
ESP32-S3-PICO-1 (SiP, needs own FCC cert)	ESP32-S3-MINI-1 (pre-certified module, 2.4 mm tall)	Saves cert cost; adds ~1.4 mm or forces side-by-side layout
HM0360	GalaxyCore GC0308 / GC032A thin module	Cheaper, lower quality, less AI support
LS013B7DH03	0.96" SSD1306 OLED (~1.4 mm)	Brighter in the dark, 10–100× more power, not silver
BHI260AP	BMI270	$3 cheaper, no on-sensor learning
BME688	Sensirion SHT45 + Bosch BMP390	No gas/AI; better temp accuracy